Vector-Valued Functions of One Real Variable

A vector-valued function of one real variable is a function $\mathbf{r}$ whose domain is a subset D of $\mathbb{R}$ and whose outputs are all in $\mathbb{R}^2$ or all in $\mathbb{R}^3$.

→ write as ordered pair

Vector-valued functions of one real variable with outputs in $\mathbb{R}^2$

We can write $\mathbf{r}$ as an ordered pair (f, g) of real-valued functions on D.

$\vec{r}(t) = (f(t), g(t))$

real number

We call f and g the "component functions" of $\mathbf{r}$.

Vector-valued functions of one real variable with outputs in $\mathbb{R}^3$

We can write $\mathbf{r}$ as an ordered triple (f, g, h) of real-valued functions on D.

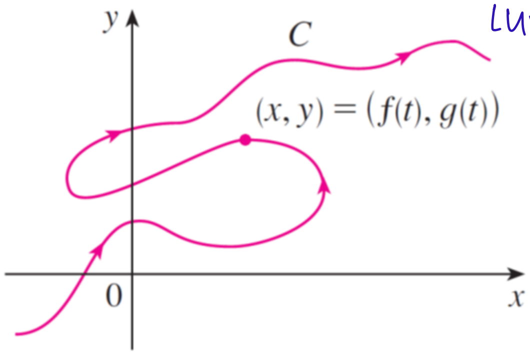
We call f, g, and h the "component functions" of $\mathbf{r}$.

$\vec{r}(t) = (f(t), g(t), h(t))$

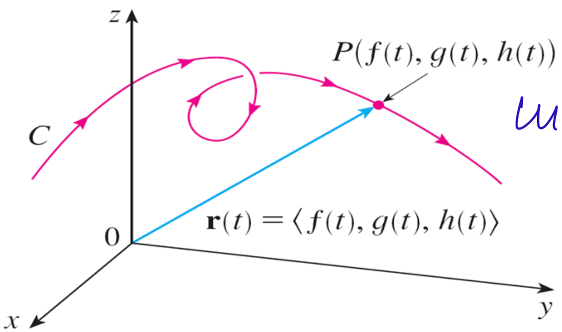
real numbers

A vector-valued function of one real variable is said to be continuous if and only if all of its component functions are continuous.

Curves and Parametrizations



curve in $\mathbb{R}^2$



curve in $\mathbb{R}^3$

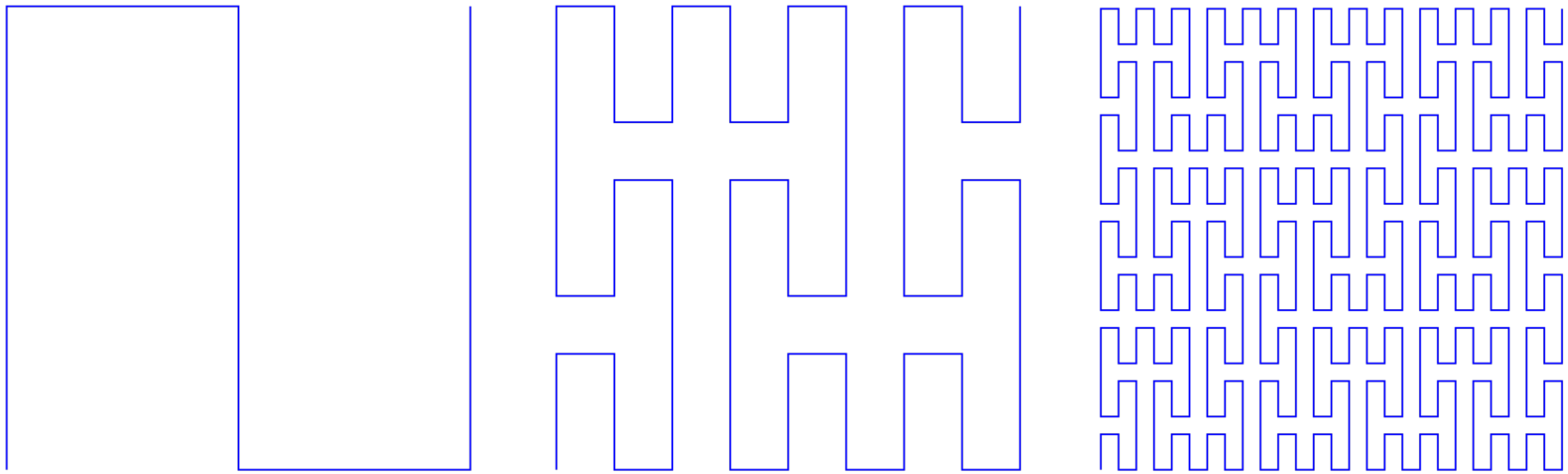
image /set of outputs

A curve in $\mathbb{R}^2$ or $\mathbb{R}^3$ is defined as the range of a continuous vector-valued function whose domain is an interval of $\mathbb{R}$ and whose outputs are in $\mathbb{R}^2$ or $\mathbb{R}^3$, respectively.

A parametrization of a curve is defined to be a continuous vector-valued function whose domain is an interval of $\mathbb{R}$ and whose range is the curve.

A Very Weird Curve

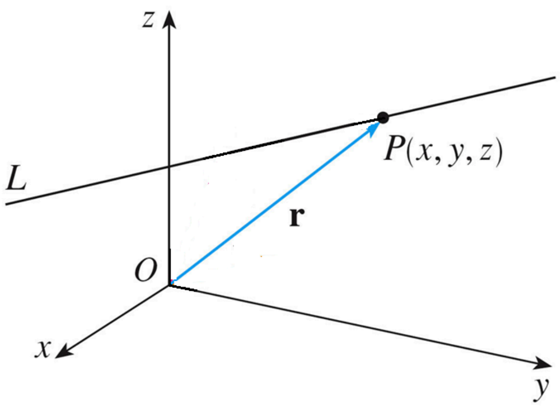
Peano curve
space-filling curve

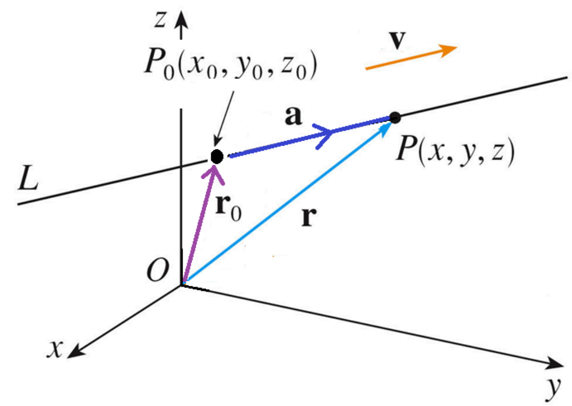


The closed unit square $[0,1]\times[0,1]$ is — believe it or not — a curve!

Guiseppe Peano (1858−1932) found a continuous vector-valued function from the interval $[0,1]$ to $\mathbb{R}^2$ whose range is the closed unit square.

Affine Parametrizations of a Line





Let L be a line in $\mathbb{R}^3$. We can parametrize L by using a single point P_0 on L and a direction vector $\mathbf{v}$ parallel to L.

A point P in $\mathbb{R}^3$ clearly belongs to L if and only if the difference between its position vector $\mathbf{r}$ and the position vector $\mathbf{r}_0$ of P_0 is parallel to $\mathbf{v}$ and hence a scalar multiple of $\mathbf{v}$:

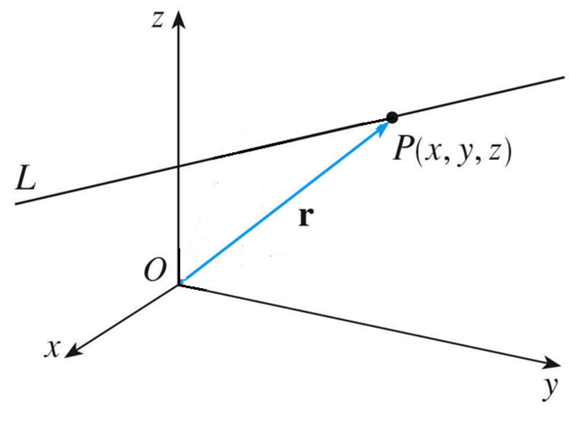
$$\underbrace{\mathbf{r}-\mathbf{r}_0}_{\overrightarrow{P_0P}} = \underbrace{t}_{\text{parameter}} \cdot \overbrace{\mathbf{v}}^{\text{fixed}}.$$

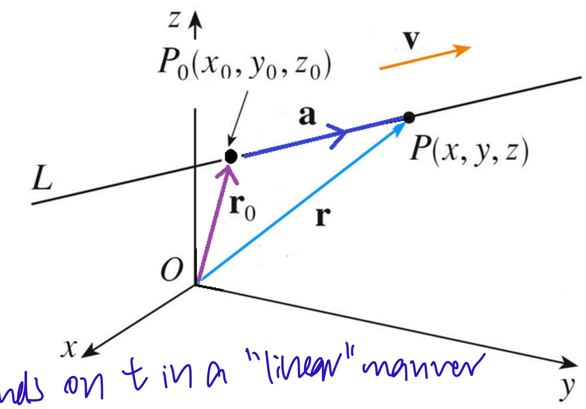
fixed

fixed

* vectors that point in opp directions are negative scalar multiples of each other

Affine Parametrizations of a Line





r(t) depends on t in a "linear" manner

Hence, the vector function $\mathbf{r}\colon \mathbb{R} \to \mathbb{R}^3$ defined by

$$\mathbf{r}(t) := \mathbf{r}_0 + t \cdot \mathbf{v}$$ → rewriting previous equation

is a parametrization of L, called an "affine parametrization" of L. By varying $\mathbf{r}_0$ and $\mathbf{v}$, we get different affine parametrizations of L.

vector equation of line L ⇒ from JC

Non-Affine Parametrizations of a Line

$\overrightarrow{AB} \,//\, L$, so $\overrightarrow{AB}$ is a direction vector

Example

Let L be the line in $\mathbb{R}^3$ containing the points $A(-1,2,0)$ and $B(3,-2,5)$.

Then the vector function $\mathbf{r}\colon \mathbb{R} \to \mathbb{R}^3$ defined by

$$\mathbf{r}(t) \coloneqq \overrightarrow{OA} + t \cdot \overrightarrow{AB} = (-1,2,0) + t \cdot (4,-4,5) = (-1+4t, 2-4t, 5t)$$

is an affine parametrization of L.

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (3,-2,5) - (-1,2,0)$
$= (4,-4,5)$

replace t by t^3

Notice that the vector function $\mathbf{s}\colon \mathbb{R} \to \mathbb{R}^3$ defined by

$$\mathbf{s}(t) \coloneqq \overrightarrow{OA} + t^3 \cdot \overrightarrow{AB} = (-1,2,0) + t^3 \cdot (4,-4,5) = (-1+4t^3, 2-4t^3, 5t^3)$$

is also a parametrization of L.

However, the <u>non-linear</u> dependence of $\mathbf{s}$ on its parameter means that it is a <u>non-affine</u> parametrization of L.

cannot replace by t^2 $\Rightarrow$ cannot get negative equation

The Standard Parametrization of a Circle

$\vec{r}(t) = \begin{cases} (t, \sqrt{1-t^2}), \text{ if } -1 \le t \le 1 \\ (t, ?), \text{ if } ?? \end{cases}$

The cosine and sine functions are perfectly suited to parametrizing circles.

The standard parametrization of the unit circle is the vector function $\mathbf{r}\colon [0,2\pi] \to \mathbb{R}^2$ defined by

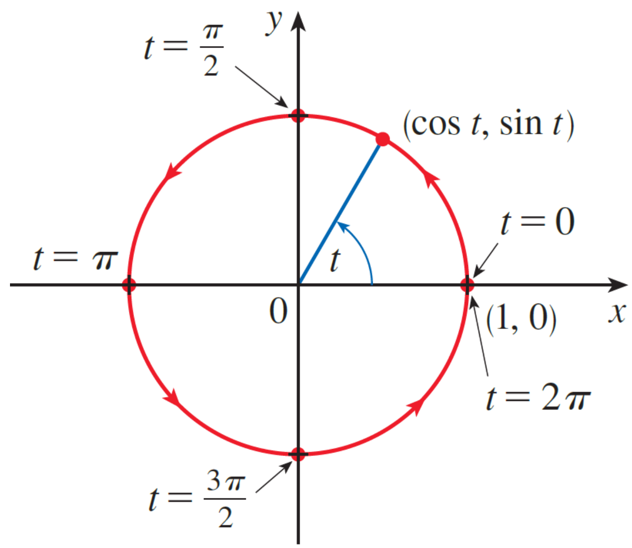
$$\mathbf{r}(t) \coloneqq (\cos(t), \sin(t)).$$

counter clockwise orientation

The standard parametrization of the circle with center (a, b) and radius r is the vector function $\mathbf{r}\colon [0,2\pi] \to \mathbb{R}^2$ defined by

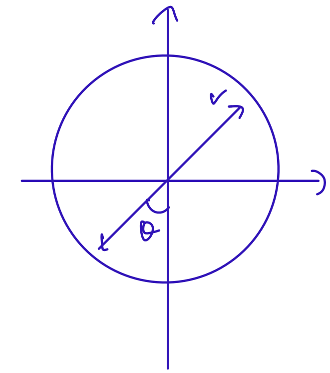
$$\mathbf{r}(t) \coloneqq (a + r\cos(t), b + r\sin(t)).$$

$= (a,b) + r \cdot (\cos(t), \sin(t))$

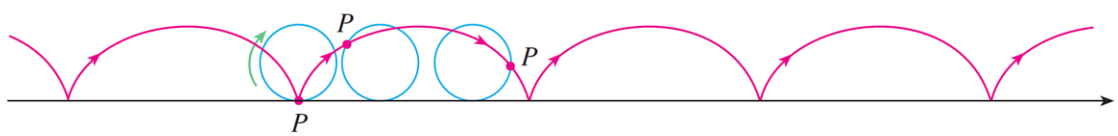


unit circle

Parametrizing a Cycloid



$(-r\sin(\theta), -r\cos(\theta))$

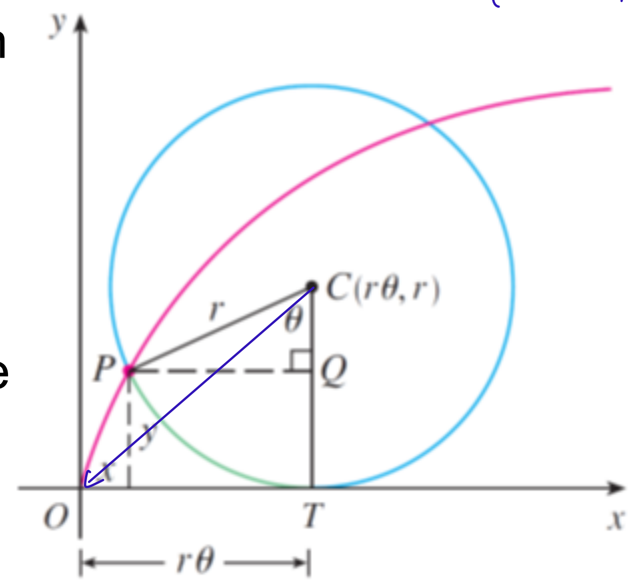


Suppose that the circle has rotated through an angle of θ:

- Its center is now at $(r\theta, r)$. Add
- If we translate the center back to the origin, then P becomes $(-r\sin(\theta), -r\cos(\theta))$.

Hence, the cycloid can be parametrized by the vector function $\mathbf{r}\colon \mathbb{R} \to \mathbb{R}^2$ defined by

$$\begin{aligned}\mathbf{r}(\theta) &\coloneqq (r\theta, r) + (-r\sin(\theta), -r\cos(\theta)) \\ &= \big(r(\theta - \sin(\theta)), r(1 - \cos(\theta))\big).\end{aligned}$$



$\sin\theta = \frac{PQ}{r}$

$PQ = r\sin\theta$

$\cos\theta = \frac{CQ}{r}$

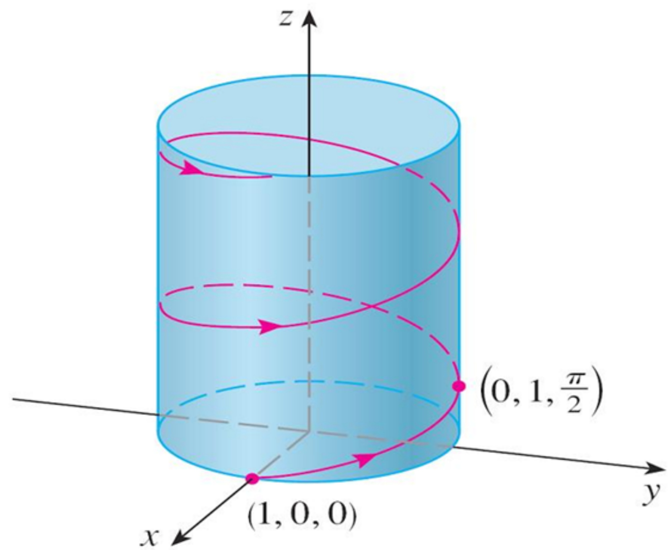
$CQ = r\cos\theta$

Parametrizing a Helix

The helix shown here can be parametrized by the vector function $\mathbf{r}\colon \mathbb{R} \to \mathbb{R}^3$ defined by

$$\mathbf{r}(\theta) \coloneqq (\cos(\theta), \sin(\theta), \theta).$$

This is because a rotation through an angle of θ results in a change in height of θ.



Deriving a Cartesian Equation for a Curve from a Parametrization

Problem

A curve C in $\mathbb{R}^2$ is parametrized by $\mathbf{r}\colon [0,2\pi] \to \mathbb{R}^2$, defined by

$$\mathbf{r}(t) \coloneqq (3\cos(t), 2\sin(t)).$$

Find a function $F\colon \mathbb{R}^2 \to \mathbb{R}$ such that $C = \{(x,y) \in \mathbb{R}^2 \mid \underbrace{F(x,y) = 0}_{\text{cartesian eq}^{\underline{n}} \text{ for } C.}\}$.

Sol$^{\underline{n}}$

- Let $(x,y) \in \mathbb{R}^2$
- Then

$x = 3\cos(t)$, $y = 2\sin(t)$ for some $t \in [0,2\pi]$

$\Leftrightarrow \frac{x}{3} = \cos(t)$, $\frac{y}{2} = \sin(t)$ for some $t \in [0,2\pi]$

$\Leftrightarrow \left(\frac{x}{3}, \frac{y}{2}\right)$ lies on the unit circle

$\Leftrightarrow \left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$ ← ellipse

$\Leftrightarrow \left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 - 1 = 0$

$F\colon \mathbb{R}^2 \to \mathbb{R}$ is defined by

$F(x,y) = \left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 - 1$

Deriving a Cartesian Equation for a Curve from a Parametrization

Problem

Parameter Space

A curve C in $\mathbb{R}^2$ is parametrized by $\mathbf{r}: \mathbb{R} \to \mathbb{R}^2$, defined by

$$\mathbf{r}(t) \coloneqq (t^2 + t, t^2 - t).$$

Find a function $F: \mathbb{R}^2 \to \mathbb{R}$ such that $C = \{(x, y) \in \mathbb{R}^2 | F(x, y) = 0\}$.

Sol^n

- Let $(x,y) \in \mathbb{R}^2$
- Then

$x = t^2 + t$, $y = t^2 - t$ for some $t \in \mathbb{R}$

$\Leftrightarrow$ $x+y = 2t^2$, $x - y = 2t$ for some $t \in \mathbb{R}$

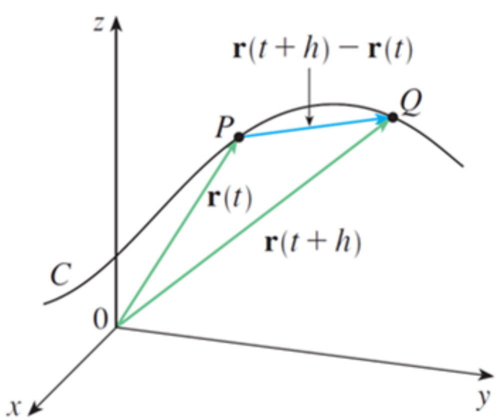
$\Leftrightarrow$ $\frac{x+y}{2} = t^2$, $\frac{x-y}{2} = t$ for some $t \in \mathbb{R}$

$\Leftrightarrow$ $\frac{x+y}{2} = \left(\frac{x-y}{2}\right)^2$

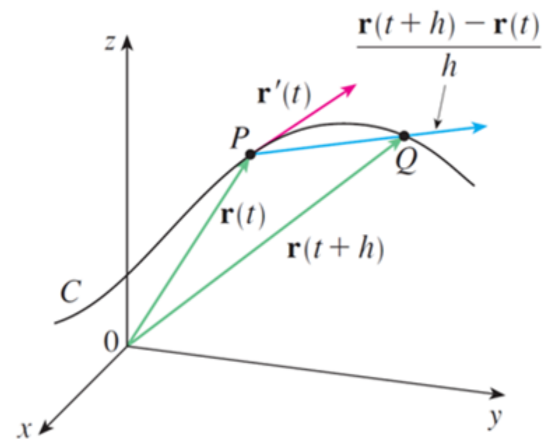
$F: \mathbb{R}^2 \to \mathbb{R}$ is defined by

$F(x,y) = \frac{x+y}{2} - \left(\frac{x-y}{2}\right)^2$

Tangent Vectors of a Curve Parametrization



$\vec{r}$ parametrizes C.



We say that $\mathbf{r}$ is <u>differentiable</u> at a parameter t if and only if the limit

$$\lim_{s\to t}\frac{1}{s-t}\cdot\big(\mathbf{r}(s)-\mathbf{r}(t)\big)$$

exists, in which case we call it the "tangent vector" of $\mathbf{r}$ at t and denote it by $\mathbf{r}'(t)$ — note that $\mathbf{r}'(t)=\big(f'(t),g'(t),h'(t)\big)$, where f, g, and h are the component functions of $\mathbf{r}$.

Singularities of a Curve Parametrization

Let C be a curve and $\mathbf{r}$ a parametrization of C.

We say that $\mathbf{r}$ is <u>singular</u> at a parameter t if and only if one of the following two conditions holds:

- $\mathbf{r}$ is not differentiable at t.
- $\mathbf{r}$ is differentiable at t but $\mathbf{r}'(t) = \mathbf{0}$. zero vector

We say that $\mathbf{r}$ is <u>regular</u> at a parameter t if and only if it is not singular at t, i.e., it is differentiable at t <u>and</u> $\mathbf{r}'(t) \neq \mathbf{0}$.

we say that $\vec{r}$ is regular iff $\vec{r}$ is regular at every parameter

Particle Motion: Velocity, Speed, Acceleration

The motion of a particle P in 2-dimensional or 3-dimensional space can be viewed as a parametrization $\mathbf{r}$ of a curve.

We call $\mathbf{r}$ the displacement function of P.

The velocity function $\mathbf{v}$ of P is defined as the derivative of the displacement function of P, namely, $\mathbf{v} = \mathbf{r}'$.

The speed function s of P is defined as the magnitude of the velocity function of P, namely, $s = \underbrace{\|\mathbf{v}\|_2}_{\text{use the pythagorean formula}} = \|\mathbf{r}'\|_2$.

The acceleration function $\mathbf{a}$ of P is defined as the derivative of the velocity function of P, namely, $\mathbf{a} = \mathbf{v}' = \mathbf{r}''$.

need to fine

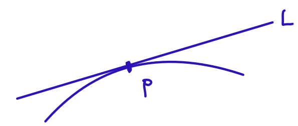
Parametrizing the Tangent Line(s) to a Curve Induced by a Parametrization of the Curve

given

Let C be a curve in $\mathbb{R}^2$ (or $\mathbb{R}^3$), P a point of C, and $\mathbf{r}$ a parametrization of C.

depends on $\vec{r}$

A tangent line to C at P induced by $\mathbf{r}$ is a line that



- passes through P and
- is parallel to the tangent vector of $\mathbf{r}$ at a parameter t whose image is P (if $\mathbf{r}$ happens to be regular at t).

Note: There could be more than one tangent line to C at P induced by $\mathbf{r}$.

used

Suppose that t is a parameter of $\mathbf{r}$ whose image is p, i.e., $\mathbf{r}(t) = P$.

Suppose that $\mathbf{r}$ is regular at t.

The line L that passes through P and is parallel to the tangent vector of $\mathbf{r}$ at t is parametrized by the vector function $\mathbf{s}\colon \mathbb{R} \to \mathbb{R}^2$ (or $\mathbf{s}\colon \mathbb{R} \to \mathbb{R}^3$) defined by

$$\mathbf{s}(\lambda) \coloneqq \overrightarrow{OP} + \lambda \cdot \mathbf{r}'(t).$$

direction vector, given by the tangent vector.

Fixed position vector

Parametrizing the Tangent Line(s) to a Curve Induced by a Parametrization of the Curve

Problem

A curve C in $\mathbb{R}^3$ is parametrized by $\mathbf{r}\colon \mathbb{R}_{>0} \to \mathbb{R}^3$, defined by

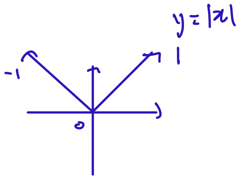
$$\mathbf{r}(t) \coloneqq (4t^3 + 5, t^2 + 2t^4, -2\ln(2t)).$$

Find a parametrization of the tangent line(s) to C at $\underbrace{(9,3,-2\ln(2))}_{\text{which parameters give rise to this}}$ induced by the parametrization $\mathbf{r}$ of C.

Sol^{n}

- $\vec{r}\,'(t) = \left(12t^2, 2t+8t^3, -\frac{2}{t}\right), \ t>0$
- $\vec{r}\,'(1) = (12, 10, -2)$
- $\vec{s}(\lambda) = (9, 3, -2\ln(2)) + \lambda \cdot (12, 10, -2)$

$$= (12\lambda + 9, 10\lambda + 3, -2\lambda - 2\ln(2)), \ \lambda \in \mathbb{R}$$



The Tangent Line to a Curve at a Point: Criteria for Existence $\underbrace{\text{(Z Z Z)}}_{\text{dangerous bend}}$

No mention of parametrization

Let C be a curve and P a point of C.

A parametrization $\mathbf{r}$ of C is said to be <u>semiregular</u> at a parameter t if and only if $\mathbf{r}$ is one-sided differentiable at t and the one-sided tangent vectors of $\mathbf{r}$ at t are non-zero.

A <u>tangent line</u> to C at P is defined as a line L with the following properties:

- We can find a parametrization $\mathbf{r}$ of C that is semiregular at all parameters whose image is P such that L contains the one-sided tangent vectors of $\mathbf{r}$ at these parameters. If you don't even have one-sided derivatives, then no tangent line.
- For <u>any other</u> parametrization $\mathbf{r}$ of C that is semiregular at all parameters whose image is P, L contains the one-sided tangent vectors of $\mathbf{r}$ at these parameters.

watch again

A tangent line may not exist, but if it exists, then it must be unique.

complicated example

A Cusp Singularity

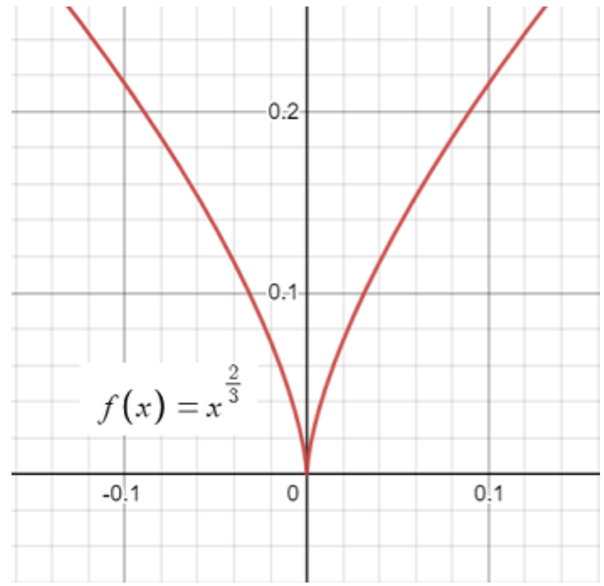
Problem

The curve C in $\mathbb{R}^2$ shown here is given by

$$C = \{(x, y) \in \mathbb{R}^2 | x^2 = y^3\}.$$

a) Show that any parametrization of C must be singular where its image is $(0,0)$.

b) Show that the y-axis is tangent to C at $(0,0)$.



A Cusp Singularity

A Cusp Singularity

$\boxed{\text{Real-Valued}}$ Functions of Two Real Variables

2 inputs, both real numbers

A real-valued function of two real variables is a function f whose domain is a subset D of $\mathbb{R}^2$ and whose outputs are real numbers.

There is only one component function, which is f itself. Why?

no. of components $\Leftrightarrow$ no. of coordinates for output

The Function Domain for an Expression

Problem

Find the domain of the real-valued function f of two real variables defined by the expression

$$\frac{x}{\sqrt{y^2 - x}}. \rightarrow \text{must be } \ge 0$$

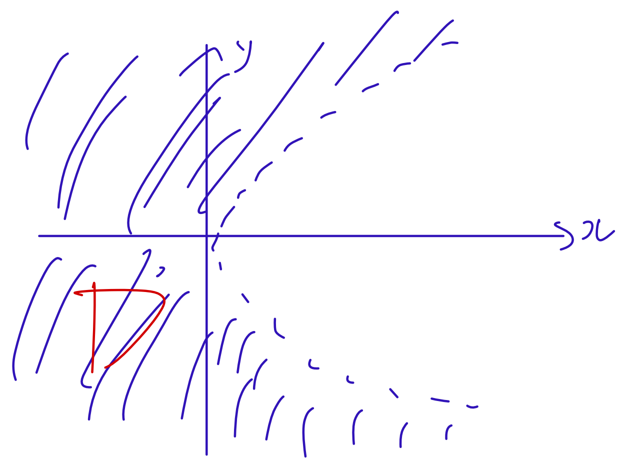
① $y^2 - x \ge 0 \iff x \le y^2$

② $\sqrt{y^2 - x} \ne 0 \iff y^2 - x \ne 0$

$\Updownarrow$

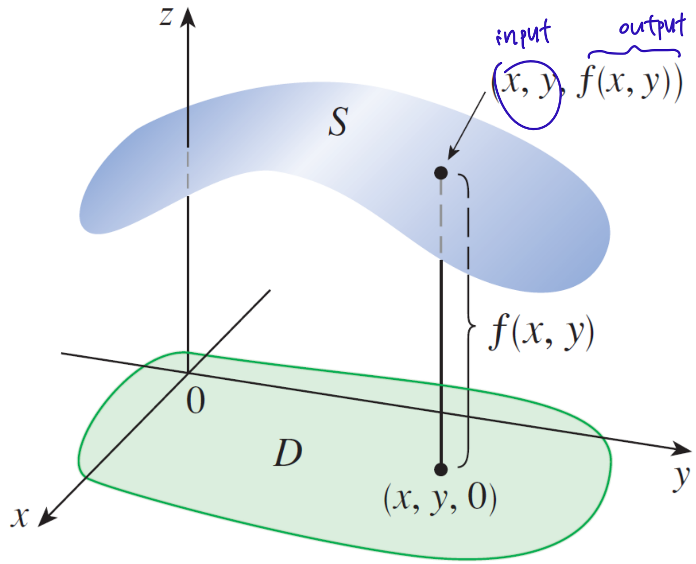
$x \ne y^2$

$\Big\}\ x < y^2$



$D = \{(x,y) \in \mathbb{R}^2 \mid x < y^2\}$

The Graph of a Real-Valued Function of Two Real Variables



Level Curves and Contour Plots

watch again

Let f be a real-valued function of two real variables with domain $D \subseteq \mathbb{R}^2$.

The level curve of f for a real number k is defined as the set of points of D that f maps to k, i.e., $\{(x, y) \in D | f(x, y) = k\}$.

A contour plot of f for a set S of real numbers is defined as the union of the level curves of f for $k \in S$.

Level Curves and Contour Plots

Problem

A function $f\colon \overbrace{\{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leq 36\}}^{\text{domain}} \to \mathbb{R}$ is defined by

$$f(x, y) \coloneqq \sqrt{36 - x^2 - y^2}.$$

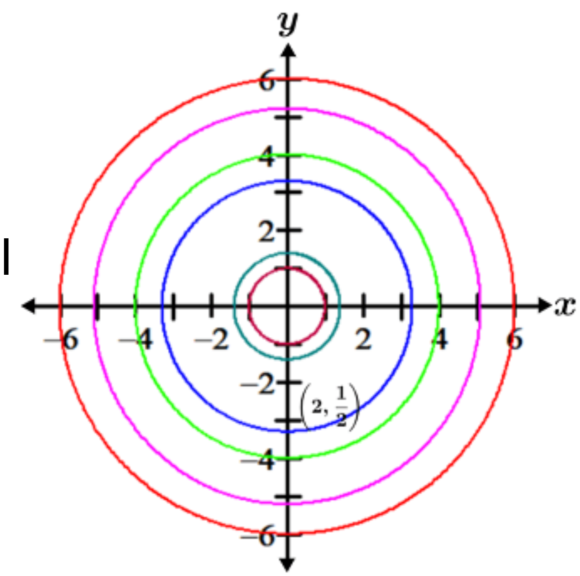
Sketch the contour plot of f with contour interval 1, going from 0 to 6.

$f(x,y) = k \iff \sqrt{36-x^2-y^2} = k$

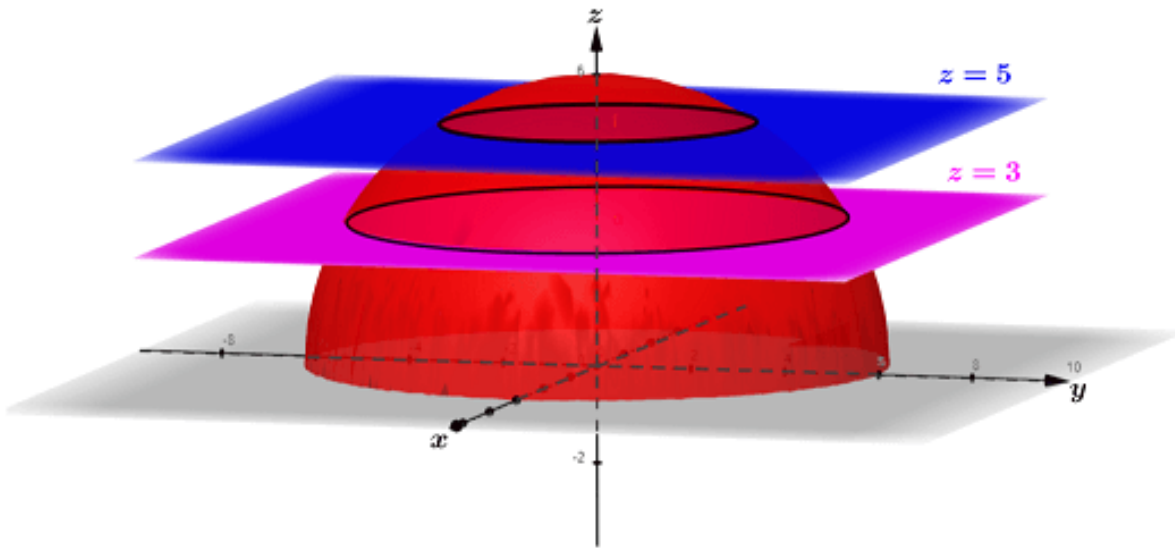
$\iff 36 - x^2 - y^2 = k^2$

$\underbrace{x^2+y^2}_{\text{circle!}} = 36 - k^2$

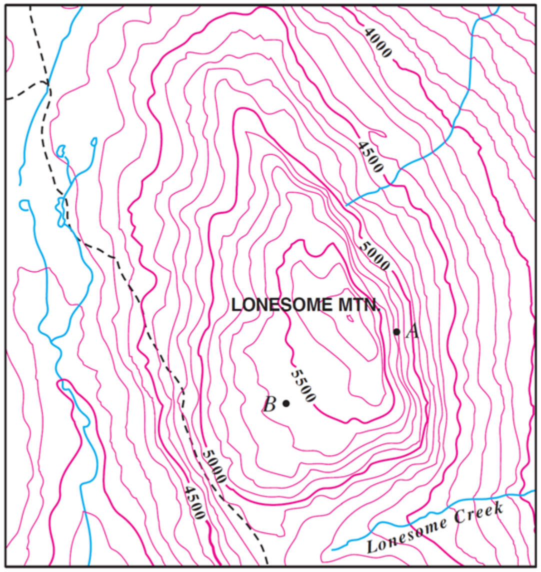
$k \in \{0,1,2,3,4,5,6\}$



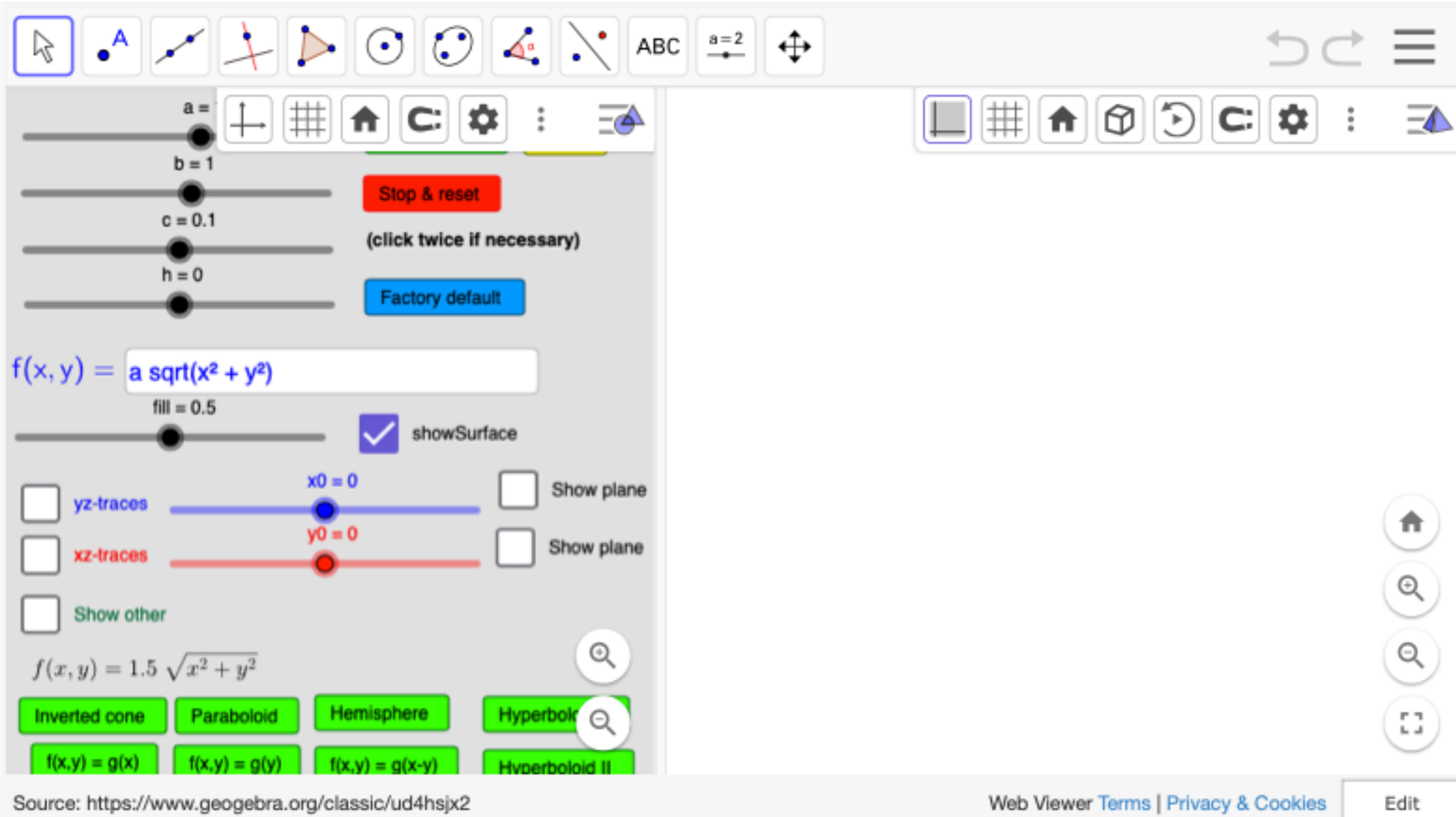
Level Curves and Contour Plots



Contour Plots and Topographic Maps



Question: What is the contour interval of this topographic map? 100


ABC
a = 2
a =
b = 1
Stop & reset
c = 0.1
(click twice if necessary)
h = 0
Factory default
f(x, y) = a sqrt(x² + y²)
fill = 0.5
showSurface
x0 = 0
yz-traces
Show plane
y0 = 0
xz-traces
Show plane
Show other
f(x, y) = 1.5 √(x² + y²)
Inverted cone
Paraboloid
Hemisphere
f(x,y) = g(x)
f(x,y) = g(y)
f(x,y) = g(x-y)
Hyperboloid II
Source: https://www.geogebra.org/classic/ud4hsjx2
Web Viewer Terms | Privacy & Cookies
Edit